

Sahil Shivaji Sawant

Shrewsbury, MA | +1 7163902316 | sahilshivajisawant@gmail.com | [LinkedIn](#) | [GitHub](#) | [Website](#)

Work Experience

University at Buffalo | *AI Research Assistant*

Jun 2026 - Present

- Integrated a Neo4j knowledge graph with a Qdrant vector store (BGE-M3 dense + BM25 hybrid search) into a single retrieval layer that runs both paths in parallel per question, allowing the university-hosted Qwen-27B model to answer queries across 450+ scraped department pages with higher relevance
- Cut median response latency from ~90s to ~45s by disabling Qwen's internal reasoning on Cypher generation while keeping it on for answer synthesis.
- Scraped and normalized HTML, Markdown, and JSONL from Playwright/Firecrawl crawls into a Neo4j graph, modeling faculties, courses, labs, research areas, and policies, which enabled students to run queries and quickly locate relevant information

Arta Support | *AI Software Engineer*

Mar 2026 - May 2026

- Scaled a troubleshooting system into an enterprise platform with ~30s response latency by orchestrating 4 AI agent workflows (router, intent, clarification, diagnosis) powered by Claude Sonnet, optimizing production performance via Anthropic prompt-caching, Socket.IO tool-calling, and conditional self-correction retries.
- Paired Pinecone with Neo4j in a hybrid graph-guided RAG system, improving retrieval precision by 30% by landing agents directly on exact knowledge-graph nodes rather than relying on standard context-stuffing.
- Automated a PDF-to-Knowledge-Graph ingestion pipeline via Text-to-Cypher query generation cutting knowledge-base update time by 40% and keeping vector indices synchronized with Neo4j.
- Deployed a full-stack AWS authentication and LLM token-accounting systems (Cognito, DynamoDB, KMS, EC2), delivering role-based dashboards, self-serve API key rotation, and per-model usage tracking to enforce cross-region budget controls.

Capgemini | *Analyst*

Dec 2022 - May 2024

- Engineered an enterprise text-to-SQL RAG system using LangGraph and Llama-2-70B over Oracle EBS schemas, enabling non-technical warehouse staff to query aerospace inventory databases in plain language.
- Slashed prompt token overhead by 65% and eliminated hallucinated column errors by abandoning static prompt definitions in favor of a dynamic retrieval mechanism that grounds generation on live, runtime SCM schema metadata.
- Improved the system to handle 5,000+ weekly inventory queries with a 94% user-acceptance accuracy rate, delivering a FastAPI backend and React frontend that stream verified SQL, raw data arrays, and generate analytical charts using Matplotlib

Projects

DeepSpeech Therapy – Real-Time Voice AI Coach | DeepSpeech Therapy – Real-Time Voice AI Coach

- Enabled speech clarity detection using a 1D convolutional autoencoder (96.89% accuracy) and a CNN + Transformer for stutter diagnosis (block, prolongation, repetition).
- Routed detected speech errors from a real-time analysis stack to a GPT-4o coaching agent that delivers context-aware articulation feedback.

DermAI - AI powered dermatology Assistant | DermAI - AI powered dermatology Assistant

- Orchestrated a multimodal dermatology assistant that routes a vision ensemble (ResNet50, DenseNet121, VGG-19, EfficientNet-B0) as a classification tool feeding a Gemini 3 Flash LLM for diagnostic feedback across 7 lesion categories on HAM10000.
- Applied SMOTE and class-weighted training to address severe class imbalance, reaching 94% precision up from 87% comparing single-model vs. ensemble output.

B.O.L.T - Quadruped Vision & Voice Control | B.O.L.T - Quadruped Vision & Voice Control

- Assembled a vision-guided object-following for the Unitree Go2 quadruped robot, running a quantized YOLOv8n model on Jetson Nano at 48ms latency with 90%+ target-tracking success indoors.
- Created a voice AI pipeline for the robot using Whisper STT, and a regex-based NLP parser dispatching commands over SSH to the on-robot execution loop supporting 20+ distinct behaviors.

Skills

Languages & Libraries: Python, PySpark, SQL/PL-SQL, TensorFlow, PyTorch, Scikit-Learn, Pandas, NumPy, SciPy, Matplotlib, Seaborn, Spacy, NLTK

GenAI & ML: Agentic Workflows, MCP, Agent to Agent communication, pydantic-ai, Lang-Graph, Lang-Chain, RAG Systems, RAG Evaluation (RAGAS), Cohere Reranking, LLM Orchestration, Prompt Engineering, Hugging Face, CNNs, RNNs, LSTMs, Transformers, Supervised & Unsupervised Learning

MLOps & Deployment: CI/CD (GitHub Actions), Model Deployment, Model Monitoring, Model Versioning, Streamlit, Docker

Backend & Frontend: FastAPI, Node.js, React, Socket.IO, REST APIs, Git, Linux/Unix

Databases & Cloud: Neo4j, Qdrant, Pinecone, DuckDB, AWS, GCP, Azure, Docker, Apache Spark, Oracle EBS

Education

University at Buffalo | *MS, Artificial Intelligence*

Buffalo, USA Aug 2024 - Dec 2025

University of Mumbai | *BE, Electronics & Telecommunication*

Mumbai, India Aug 2018 - May 2022